Joins in R

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Introduction

- It’s rare that a data analysis involves only a single data frame.
- Typically you have many data frames, and you must join them together to answer the questions that you’re interested in.
- To understand joins, you need to first understand how two tables can be connected through a pair of keys, within each table.

Types of Keys

Primary Key

A primary key is a variable or set of variables that uniquely identifies each observation.

Compound Key

When more than one variable is needed, the key is called a compound key.

Foreign Key

A foreign key is a variable (or set of variables) that corresponds to a primary key in another table.

Ways to join data frames

- Depending on the variables and information we want to include in the resulting data frame.

- The package `dplyr` has several functions for joining data, and these functions fall into two categories, mutating joins and filtering joins.

  i. **Mutating joins** add new variables from one table to matching observations in another table.

  ii. **Filtering joins** retain observations in one table based on whether or not they match the observations in another table.

Mutating Joins

- There are four types of mutating joins:

Types of mutating joins

i. Left joins

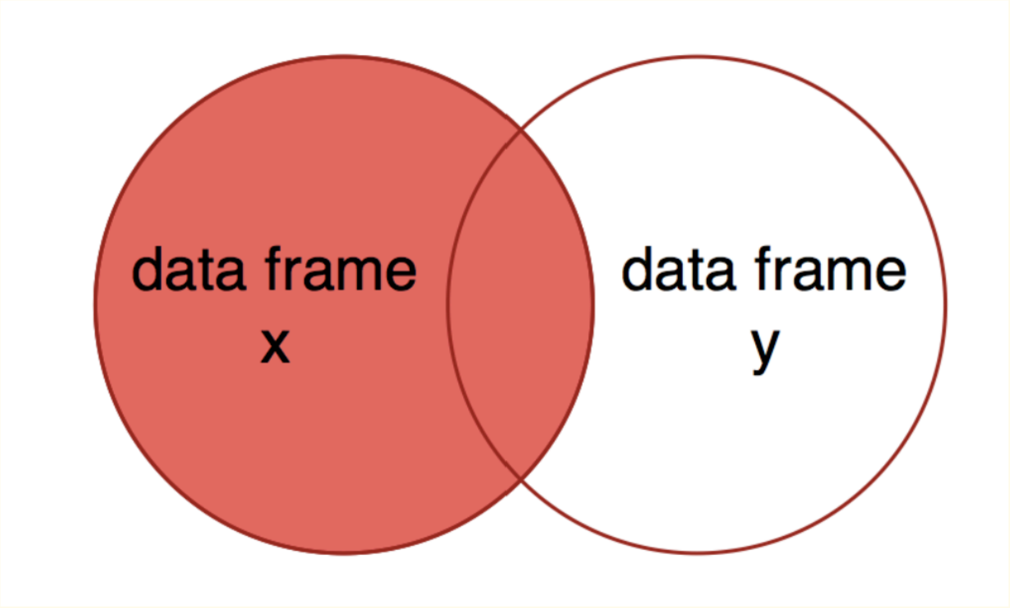
ii. Right Joins

iii. Inner joins

iv. Full joins

Left Joins: `left_join`

- The circle on the left is data frame x, and the one on the right is data frame y.
- The overlap between the two circles represents the observations with keys that are present in both data frames.
- The result of a left join is all of data frame x, plus the parts of data frame y with overlapping keys.



Example

- Consider the first data frame

Treatment	Replicate	Ammonium	Nitrate
1	1	8.2	1.7
1	2	6.9	3.6
2	1	12.1	2.8
2	2	10.5	0.8

Example

- Consider the second data frame

Treatment	Replicate	Carbon
1	1	42.5
1	2	49.1
2	1	40.8
2	2	50.4

- We are going to demonstrate how to join the two data frames

`left_join()` in R

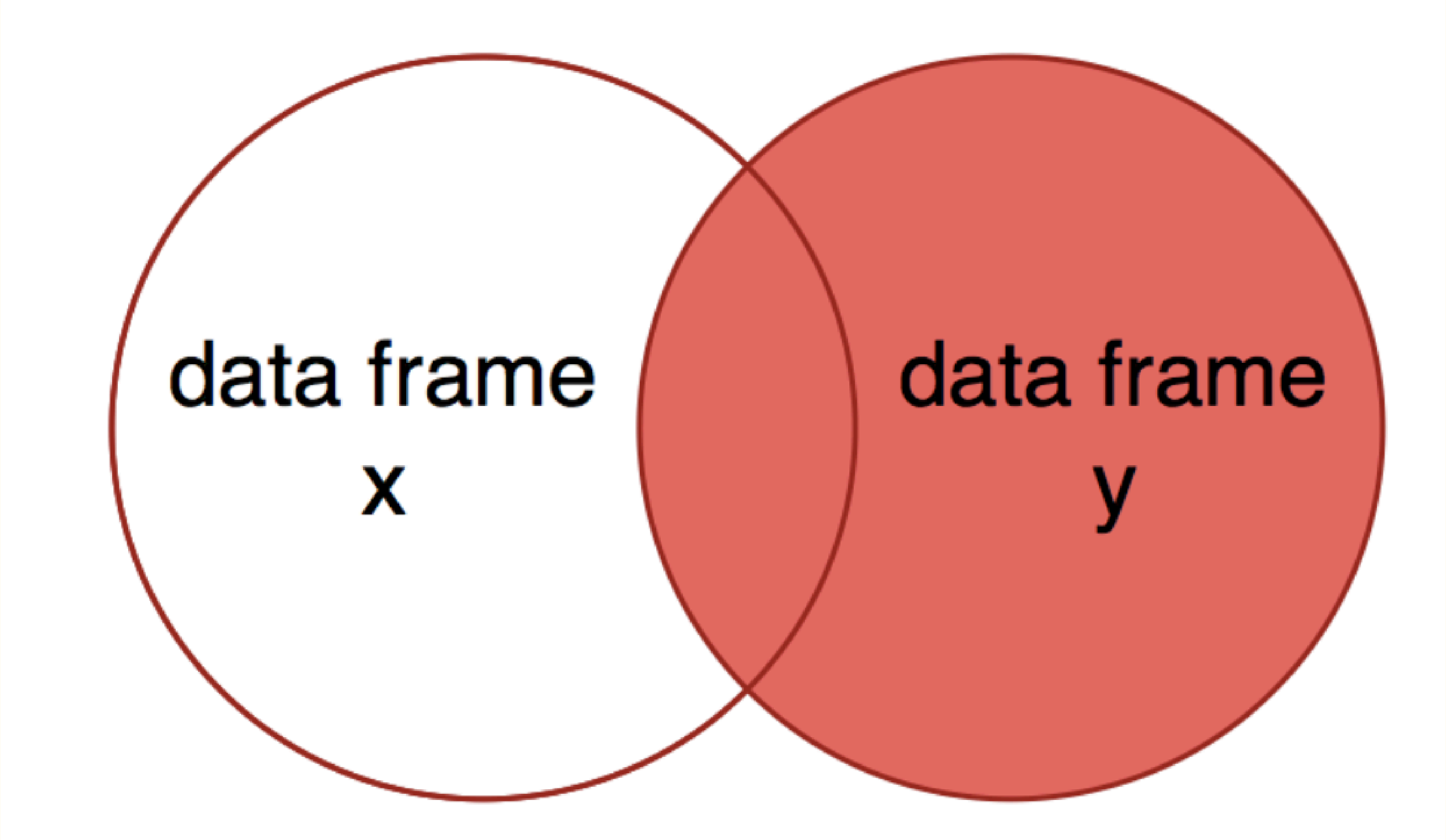
```r
# data 1
data1 <- data.frame(
  Treatment = c(1,1,2,2),
  Replicate = c(1,2,1,2),
  Ammonium = c(8.2,6.9,12.1,10.5),
  Nitrate = c(1.7,3.6,2.8,0.8)
)
# data 2
data2 <- data.frame(
  Treatment = c(1,1,2,2),
  Replicate = c(1,2,1,2),
  Carbon = c(42.5,49.1,40.8,50.4)
)
# joining the data frames
data1 %>%
  left_join(y=data2, by=c("Treatment","Replicate"))
```

Output

Treatment	Replicate	Ammonium	Nitrate	Carbon
1	1	8.2	1.7	42.5
1	2	6.9	3.6	49.1
2	1	12.1	2.8	40.8
2	2	10.5	0.8	50.4

Right Join: `right_join()`

- A right join is conceptually similar to a left join, but includes all the observations of data frame y and matching observations in data frame x - the right side of the Venn diagram.




```
# joining the data frames
data_join2 <- data_join2 %>%
  right_join(x = data1,y=data2, by=c("Treatment","Replicate"))
```

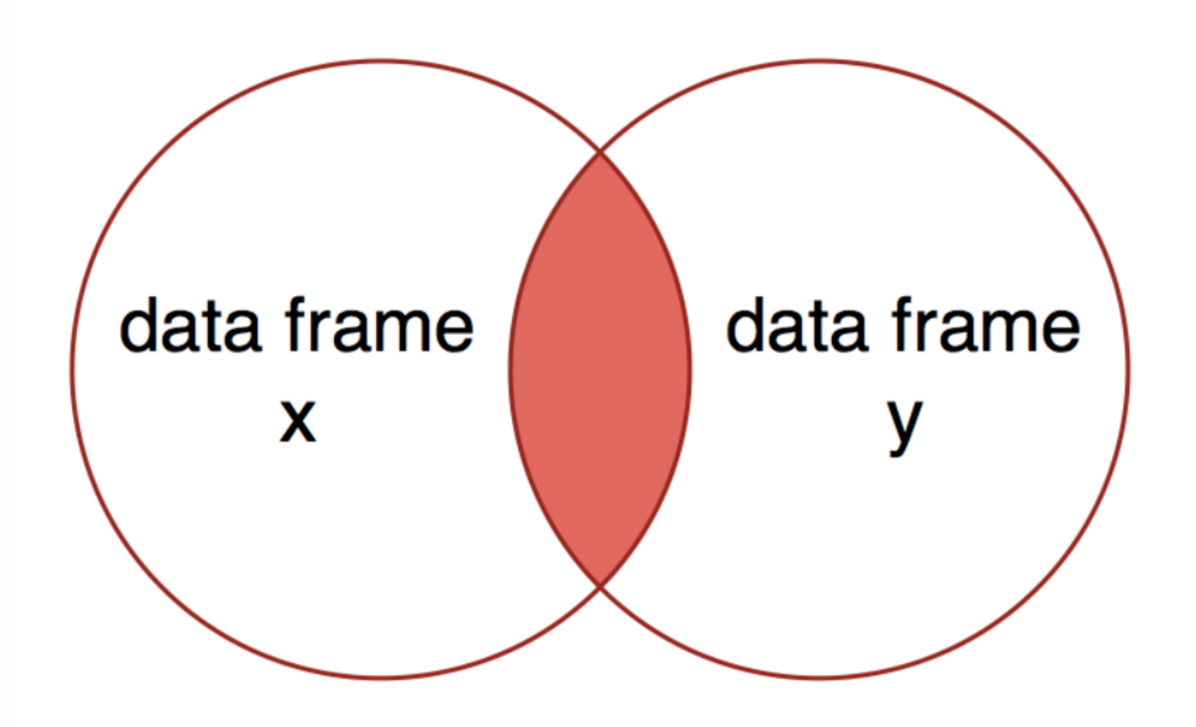
Output

Treatment	Replicate	Ammonium	Nitrate	Carbon
1	1	8.2	1.7	42.5
1	2	6.9	3.6	49.1
2	1	12.1	2.8	40.8
2	2	10.5	0.8	50.4
3	1	NA	NA	89.0
3	2	NA	NA	67.0

Inner Join `inner_join`

- An inner join includes observations with keys that are present in both data frames.

- This is the same as keeping only the observations in x that have a matching observation in y.




```r
# joining the data frames
data_join3 <- data_join3 %>%
  inner_join(x = data1,y=data2, by=c("Treatment","Replicate"))
```

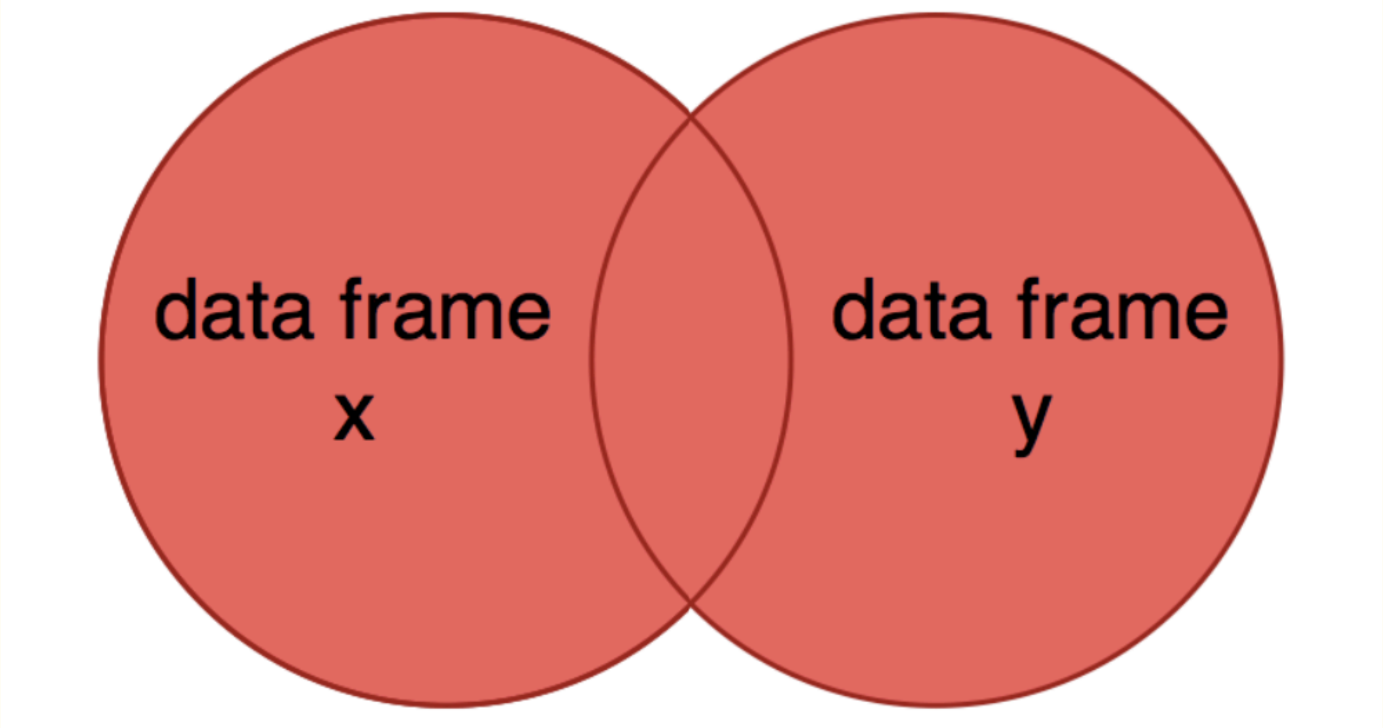
Output

Treatment	Replicate	Ammonium	Nitrate	Carbon
1	1	8.2	1.7	42.5
1	2	6.9	3.6	49.1
2	1	12.1	2.8	40.8
2	2	10.5	0.8	50.4

Full Join: `full_join()`

- You might want a data frame that includes all data from both data sets, whether or not observations are missing in one or the other.

- This is analogous to including both circles in a Venn diagram.




```r
# joining the data frames
data_join4 <- data_join4 %>%
  inner_join(x = data1,y=data2, by=c("Treatment","Replicate"))
```

Output

Treatment	Replicate	Ammonium	Nitrate	Carbon
1	1	8.2	1.7	42.5
1	2	6.9	3.6	49.1
2	1	12.1	2.8	40.8
2	2	10.5	0.8	50.4
3	1	NA	NA	89.0
3	2	NA	NA	67.0

Filtering Joins

- Filtering joins keep only observations from the first data frame, and compare observations to a second data frame to determine which observations to keep.

- Filtering joins will only ever remove observations, and never add them.

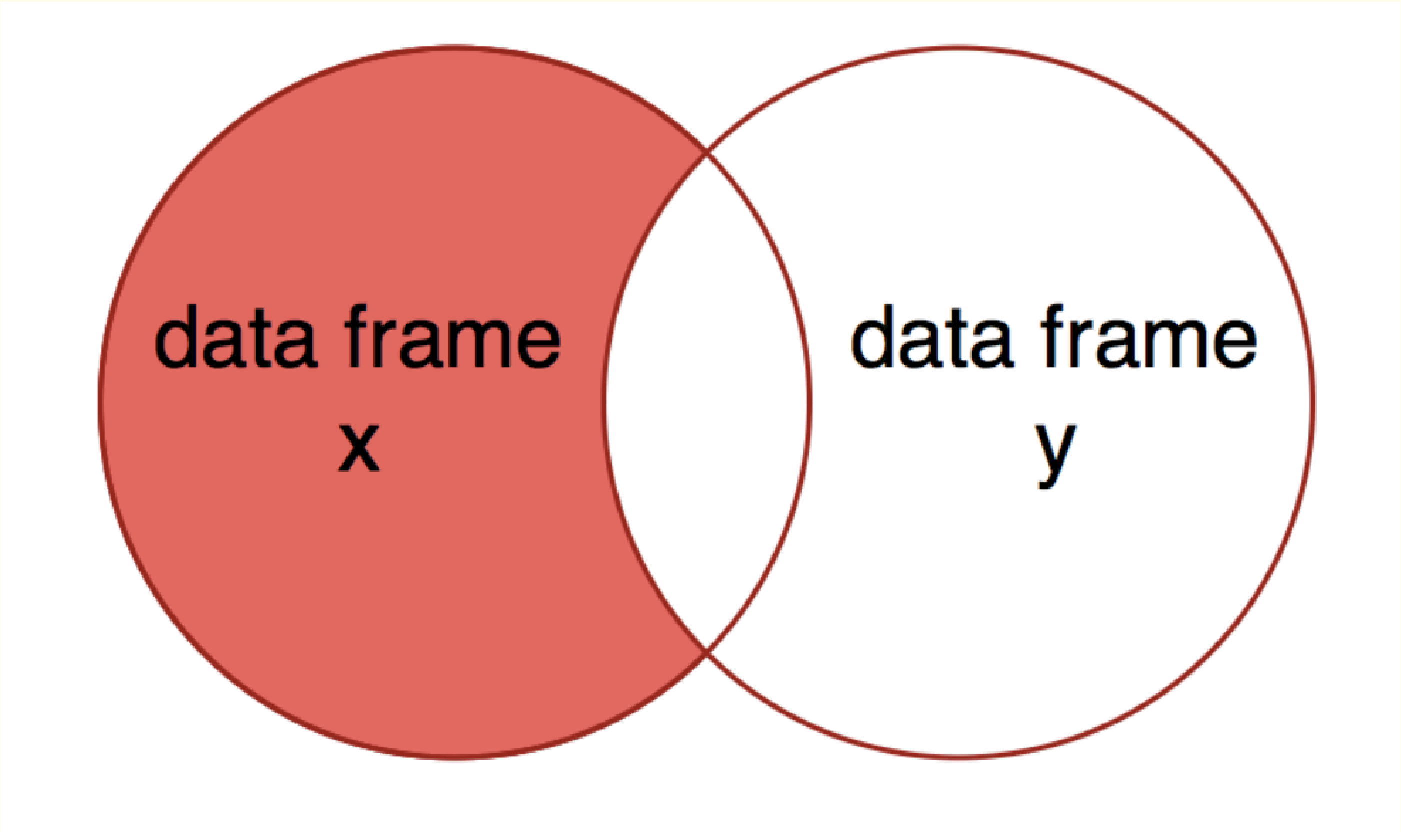
- There are two types of filtering joins:

  i. Semi joins `semi_join` - keeps all the observations in `x` that have a match in `y`

  ii. Anti joins `anti_join` - kepp all observations in `x` that do not match in `y`

Semi Join

- Semi joins keep all observations in x that have a match in y.
- The Venn diagram depicting this join is the same as that for an `inner_join`.




```r
# joining the data frames
data_join5 <- data_join5 %>%
  anti_join(x = data1,y=data2, by=c("Treatment","Replicate"))
```

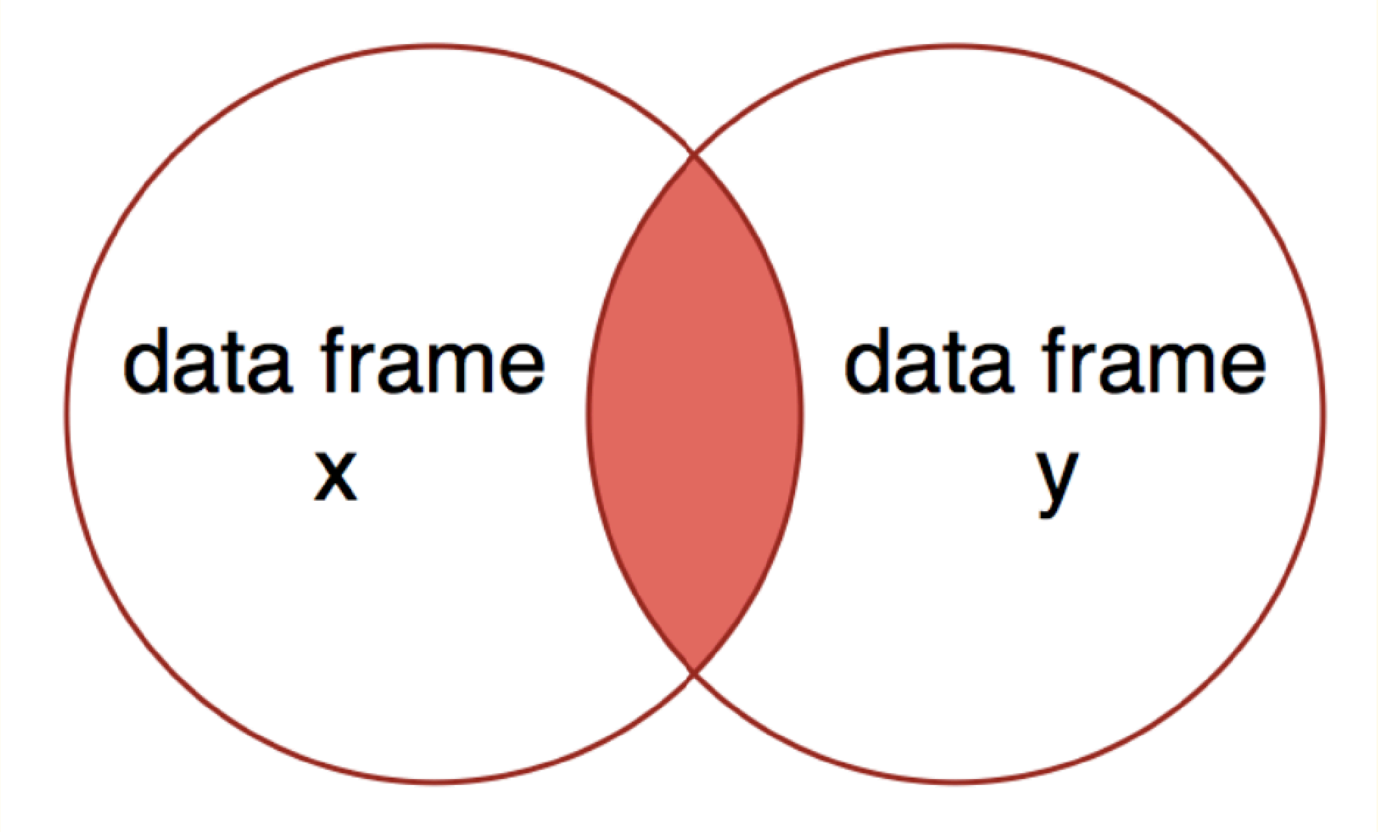
Output

Treatment	Replicate	Ammonium	Nitrate
1	1	8.2	1.7
1	2	6.9	3.6
2	1	12.1	2.8
2	2	10.5	0.8

Anti Joins

- Anti joins keep all observations in x that do not have a match in y.

- An anti join can be used to determine which observations in x are missing data in y.




```r
# joining the data frames
data_join6 <- data_join6 %>%
  anti_join(x = data2,y=data1, by=c("Treatment","Replicate"))
```

Output

Treatment	Replicate	Carbon
3	1	89
3	2	67